INORGANIC CUME
April 20, 2007

1. Using the 18-electron rule as a guide, indicate the probable number of carbonyl ligands in (a) $W(\eta^6\text{-}C_6H_6)(CO)_n$, (b) $Rh(\eta^5\text{-}C_5H_5)(CO)_n$ and (c) $Ru_3(CO)_n$.

2. Propose two syntheses for $MnMe(CO)_5$, both starting with $Mn_2(CO)_{10}$, with one using Na and one using Br_2. You may use other reagents of your choice.

3. Provide plausible reasons for the differences in IR wavenumbers between each of the following pairs: (a) $Mo(PF_3)_3(CO)_3$ 2040, 1991 cm^{-1} versus $Mo(PMe_3)_3(CO)_3$ 1945, 1851 cm^{-1}, (b) $MnCp(CO)_3$ 2023, 1939 cm^{-1} versus MnCp2*$(CO)_3$ 2017, 1928 cm^{-1}.

4. Which metal carbonyl in each of (a) $[Fe(CO)_4]^{2-}$ or $[Co(CO)_4]^-$, (b) $[Mn(CO)_5]^-$ or $[Re(CO)_5]^-$ should be the most basic towards a proton? What are the trends upon which your answer is based?

5. Suggest a sequence of reactions for the preparation of $Fe(dppe)(CO)_3$, given iron metal, CO, dppe ($Ph_2PCH_2CH_2PPh_2$), and other reagents of your choice.